

Persistent Monitoring and Analysis at a Corridor Scale with Lidar

William Barbour*, Matthew Bunting*, Derek Gloudemans*,
Daniel B. Work*, Jonathan Sprinkle*

*Institute for Software Integrated Systems, Vanderbilt University, Nashville, TN, USA

{william.w.barbour, matthew.r.bunting, derek.gloudemans, dan.work, jonathan.sprinkle}@vanderbilt.edu

Abstract—This paper describes the persistent monitoring with mounted lidar sensors of a transportation corridor over the course of 120 uninterrupted days in Nashville, TN. The data for these 120 days are publicly released as part of publication of this work. The goal of this monitoring is to provide opportunities to design interventions for unsafe intersections, better understand traffic dynamics, and explore the potential for closed-loop control of vehicle signals and walk signs. Lidar units are installed with some overlapping, and some non-overlapping fields of view, offering coverage of eight locations (five intersections and three mid-block areas) across a span of two miles. Continuous object trajectories for all modes of travel are generated from edge processing of raw lidar point cloud data. Unlike camera-based sensing, lidar is natively privacy preserving. This provides an opportunity for improved reception in communities. The paper provides a full description of the corridor, the types of classifications performed by each lidar installation, and the refresh rates and data types recorded. In addition, sample analyses are given to demonstrate the richness of the data. Sample results include hot spots for near-miss events between classified objects, daily turning count statistical analysis, and out of crosswalk pedestrian activity.

Index Terms—Lidar, persistent monitoring, post encroachment time.

I. INTRODUCTION

Light detection and ranging (LiDAR; lidar) uses time of flight of emitted laser beams in order to estimate the range and reflectivity of nearby objects. Lidar is common among high-end fully autonomous vehicles, and has been used extensively in research for self-driving and vehicle control. The use of lidar in infrastructure monitoring has grown over the past decade as the cost per unit has decreased, and as edge-based classification and tracking algorithms have improved. In particular, edge-based methods make it possible to dramatically reduce the amount of transmitted and stored data, by depending on the edge compute to identify bounding boxes, rather than storing or transmitting point cloud information.

Persistent monitoring of traffic provides a unique opportunity to observe rare events both before and after they occur, with a variety of sensing modalities used to perform the monitoring (see for example the recent review articles for sensing for vulnerable road users [1], [2]). While any autonomous vehicles in operation on streets may be continuously monitoring their environment, it is not clear that continuous monitoring while in motion at the speed of a car could capture pedestrian, cyclist, or other dynamics at a resolution and

time window (before and after an event) to establish whether interventions could take place to improve safety.

Our work mounts multiple lidar units in a fixed location along a high-injury network in Nashville, TN. Sensors are placed at intersections and mid-blocks where injuries have historically been high. Through this method, unsafe events will continue to be detected, including the before and after portion of those events. Analysis of the data in future research can determine whether near-miss events and detections can be tied to physical locations or times when unsafe events take place.

The contribution of this paper is the collection of these 120 days of data, release of the data, and documentation of their use by the community to explore mobility and safety research questions. The paper begins with context from the current state of the art for persistent monitoring of individual trajectories in Section II. Section III provides the context of the corridor on which the sensors are deployed, including the sensor locations and where overlaps exist. Section IV describes the recorded data: types, frame rates, and any other metadata. Section V provides summative results of the dataset itself that can be immediately calculated from each intersection, providing the scope of the data. Section VI demonstrates methods of analysis that are possible based on the data, including research questions that can be answered. Section VII provides information for how others may access the data, and general information regarding its size by day or event.

II. RELATED WORK

A variety of efforts to provide persistent monitoring of mobility scenarios have relied on camera sensors. Perhaps most well-known, the NGSIM dataset monitored stretches of roadway for 15-minute stretches of time using an array of cameras mounted high on fixed infrastructure [3]. The I-24 MOTION project [4] builds upon this project with modern software and hardware infrastructure, providing highway vehicle trajectories across 4.2 miles for 4.5 hours each day. More recently, the development of lightweight cameras and capable drones have enabled persistent monitoring. The HighD, [5], ExiD [6], AUTOMATUM [7], HIGH-SIM [8], and the *Third Generation Simulation* (TGSIM) [9] datasets use either drone or helicopter-mounted cameras to produce primarily highway vehicle trajectory data of up to 2 hours in duration. Likewise,

pNEUMA [10], OpenDD [11], roundD [12], inD [13], Interaction [14] and CitySim [15] datasets utilize drones or swarms of drones to study vehicle and pedestrian interactions in urban settings such as roundabouts and intersections.

Most recently, the Songdo Traffic dataset leveraged a swarm of 20 drones to capture synchronized mobility trajectory data across 20 intersections for 30 minutes at a time. Over 700,000 trajectories are captured at 30 Hz [16]. High aerial fields of view make modern image segmentation algorithms [17] well posed for vehicle tracking in these contexts; thus in general drone-based methods yield greater flexibility over fixed infrastructure sensors for monitoring multiple roadways at the cost of relatively short battery life and the need for human piloting. Further, the work [18] recently demonstrated through a case study of the multi modal intersection in Salford, Manchester, the ability of short drone flights to estimate critical intersection features across the day.

With the advance of lidar hardware at a dramatically lower cost compared to a few years ago, there is a growing interest to permanently mount lidar at intersections to monitor traffic and vulnerable road users. For example, the work [19] installed lidar at an intersection and reported PET values as determined by the sensor and identified 857 vehicle-vehicle conflicts via post encroachment time over a five month period. The most closely related work to the present article is TUM Traffic (TUMTraf) Intersection Dataset [20], which consists of labeled LiDAR point clouds and synchronized camera images.

III. CORRIDOR DESCRIPTION

This section describes the corridor from a transportation context, as well as how the sensors are deployed on it, with overlapping coverage of those sensors pointed out.

The corridor is located in the North Nashville community. The project corridor is located roughly two miles from downtown Nashville and contains predominately single- and multi-family residential housing and commercial businesses. The roughly two-mile project corridor starts at the intersection of Clarksville Pike and 26th Avenue North and travels southeast along Clarksville Pike to Dr. D.B. Todd Junior Boulevard. The project corridor then turns east onto Buchanan Street and ends at the intersection of Buchanan and 9th Street North.

The corridor contains five signalized intersections (Location 1: 26th and Clarksville; Location 2: 25th and Clarksville; Location 5: DB Todd and Clarksville; Location 6: DB Todd and Buchanan; and Location 8: Buchanan and 9th). Within the corridor, three additional un-signalized intersections are also equipped (Location 3: Midblock 1; Location 4: Midblock 2; Location 7: Midblock 3), resulting in a total of eight installation locations. The eight locations are displayed in Figure 1. The installation along this corridor was selected due to its presence on the Nashville Department of Transportation (NDOT) high injury network, i.e., portions of the road network with a disproportionate number of injuries due to crashes.

A typical signalized intersection installation is shown in Figure 2. The Lidar devices are Ouster OS1 units, which have 128 vertical channels operating at 10 Hz with an effective



Fig. 1. Location of the corridor in Nashville, TN instrumented with lidar sensors. Yellow circles indicate the 80 m range of each sensor, located at the centerpoint of the sensor.



Fig. 2. Sensors are typically installed in pairs, on opposite sides of an intersection connected with power over ethernet. An edge compute box receives data from each sensor and produces object detections and tracks.

range of 80m. The sensors are installed in pairs on signalized intersections and connect to an edge computer for object detection and tracking. The edge computer is a GPU-equipped embedded computer provided by the lidar manufacturer, which features an NVIDIA Jetson AGX Orin module. The edge computer unifies the data from both lidar units (the typical signalized intersection installation) and performs detection of the objects in the scene. It also provides power to the sensor via power over ethernet and has the capability to communicate with the traffic signal controller (not configured in our installation) and an external cellular modem (configured in our installation) when desired.

IV. DATA AND CLASSIFICATION

The data used in this study is derived from a series of roadside sensor deployments at intersections across North Nashville. Each deployment collects high-resolution trajectory data of road users—including vehicles, pedestrians, and bicycles—24 hours a day. Data is streamed from the sensors to cloud storage using cellular backhaul. This enables flexible deployment without the need for fiber infrastructure, though

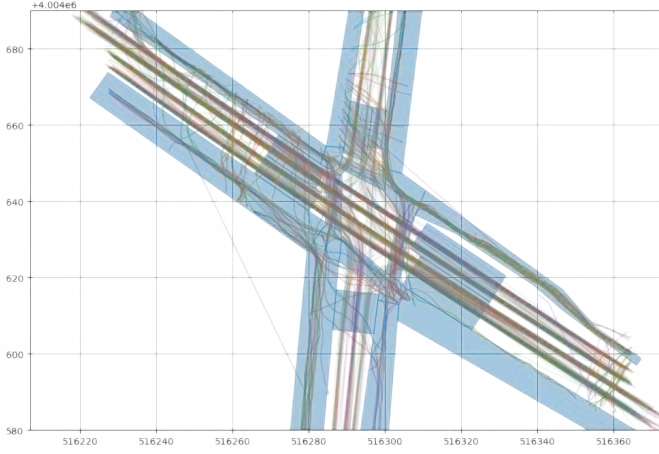


Fig. 3. Trajectories on a single intersection across one day, all modes of travel, for October 16, 2024 (location 1).

occasional network disruptions can cause missing data or irregular sampling intervals. These disruptions are typically reflected in the time-series data as timestamp gaps or varying frame rates and should be accounted for in downstream analyses.

Each file included in the dataset captures the motion of all detected objects within a particular sensing period. For example, the file 20241212-2.pk1 contains 48,695 object tracks recorded continuously between 00:00:00 and 23:59:59 on December 12, 2024 (location 2). Each object is stored as a dictionary containing 16 fields. These include identifiers (object_id, intersection_id), classification labels (classification, sub_classification), and estimated physical dimensions averaged over the trajectory (obj_length, obj_width, obj_height). The remaining fields consist of timestamped sequences representing the object’s motion over time. These include position (obj_x, obj_y), velocity (vel_x, vel_y), heading, and projected UTM coordinates (utm_x, utm_y), all sampled at high frequency (typically 10 Hz).

Objects are classified into subcategories including cars (38,469 for day of 2024-12-12), trucks (7,042), pedestrians (1,848), buses (613), trailers (380), and bicycles (224), with 119 unclassified entries. While GPS coordinates are not directly included, each object is tagged with an intersection_id that maps to a known physical location. The structure of the data supports a wide range of transportation research applications, including trajectory prediction, multimodal safety analysis, and operational performance benchmarking. A detailed description of all fields, including units, is provided in Table I.

V. RESULTS

A total of 13,131,884 filtered trajectories are counted throughout the 120 day duration of the dataset across all eight locations. Classification filters applied for the results include filtering for all vehicles (classified either as a ‘Vehicle’ or ‘Large Vehicle’), bicycles, and persons. Vehicles are further

filtered to ensure that the object’s length, width, and height, sum to 6.5m or larger. Such filtering reduces some outliers. In total, 12,392,172 vehicle trajectories, 91,741 bicycle trajectories, and 647,971 person trajectories, are recorded in the dataset.

Further filtering is applied for purposes of ensuring that a trajectory is of sufficient length for analysis. Within the released data, the threshold is that the distance between the first and final timestamp of a trajectory should be greater than some value, namely:

$$\sqrt{(x_0 - x_n)^2 + (y_0 - y_n)^2} \geq d$$

Where x_0, y_0 are the beginning UTM coordinates, and x_n, y_n are the final UTM coordinates. The filtering criteria for distance for each time are as follows: Vehicle, $d = 12m$; Bicycle, $d = 12m$; and Person, $d = 10m$.

Figure 4 shows, by day of the week, and by intersection, the ranges of vehicles observed. Persons and bicycles are omitted for brevity and clarity, but are available for exploration in the dataset tutorial available at [21].

Over these 120 days with 8 locations, there were outages of a total of 5 days for a single sensor (location 5), in which the sensor was offline from September 24–September 29, 2024. During this time, data from the remaining locations were still recorded. The offline data results are not included in Figure 4, to avoid outliers in the figure.

VI. ANALYSIS

With data described in the results, this section provides an approach to utilize those data for some sample analyses. The examples considered are 1) Post Encroachment Time (PET), which may indicate near collision; 2) pedestrian crossings outside crosswalks; and 3) turning movement counts from one intersection.

A. Post Encroachment Time

Unique identifiers for objects permit calculation of Post Encroachment Time (PET) [22] at each intersection. PET is a widely used surrogate safety metric that quantifies the temporal proximity between two road users who occupy the same spatial region at different times. It is particularly valuable for identifying near-miss events and evaluating potential conflict risk without requiring actual collisions.

Let two road users (e.g., vehicles) be observed over a discrete time domain. Their positions and orientations over time are represented as sequences of timestamped bounding boxes in the two-dimensional plane. Specifically, let $\{t_k\}_{k=1}^N$ and $\{t_\ell\}_{\ell=1}^M$ denote discrete time indices corresponding to the trajectories of two road users (e.g., vehicle, pedestrian, etc.). The occupancy region (bounding box) of vehicle/object 1 at time t_k is denoted as $B_1(t_k) \subset \mathbb{R}^2$, $B_2(t_\ell) \subset \mathbb{R}^2$: the occupancy region of vehicle 2 at time t_ℓ .

Each bounding box is assumed to be a closed, rectangular region derived from the vehicle’s center position, dimensions, and heading.

TABLE I
DESCRIPTION OF DATA FIELDS IN OBJECT TRAJECTORY RECORDS

Field	Units / Type	Description
object_id	Integer	Unique identifier for each tracked object.
intersection_id	Integer	Identifier for the sensor location; can be mapped to a physical intersection.
classification	String	General object type (e.g., VEHICLE, PEDESTRIAN).
sub_classification	String	Specific object type (e.g., car, truck, bicycle).
obj_length	Meters (float)	Estimated object length, derived from the bounding box.
obj_width	Meters (float)	Estimated object width.
obj_height	Meters (float)	Estimated object height.
ts	List of datetime	Timestamps for each observation in ISO format with timezone.
obj_x	Meters (List[float])	Local X coordinate of the object over time.
obj_y	Meters (List[float])	Local Y coordinate of the object over time.
vel_x	m/s (List[float])	Velocity component in the X direction.
vel_y	m/s (List[float])	Velocity component in the Y direction.
heading	Degrees (List[float])	Raw heading angle, clockwise from north.
utm_x	Meters (List[float])	UTM-projected X coordinate (Zone 16N).
utm_y	Meters (List[float])	UTM-projected Y coordinate (Zone 16N).
rot_heading	Degrees (List[float])	Heading angle adjusted for visualization.

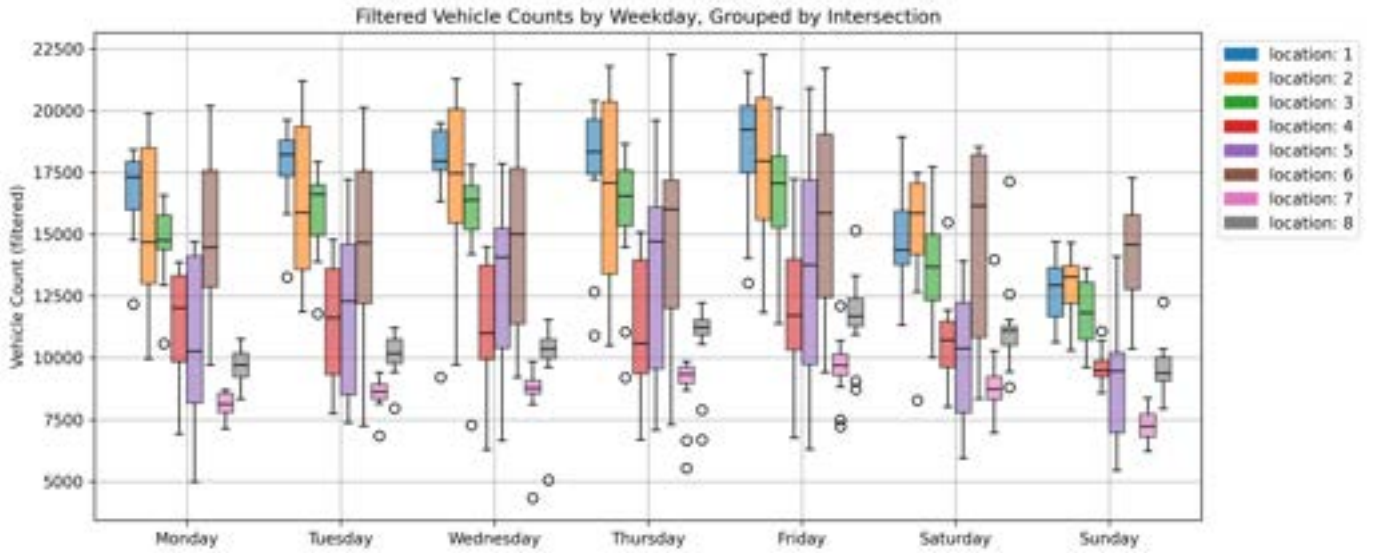


Fig. 4. All filtered vehicle counts across 120 days, with distribution across weekdays. Days where an intersection was offline are removed as outliers for plotting. Boxplots display interquartile ranges for each value; outliers more than 1.5 times the interquartile range are shown by circles.

We define the PET between two trajectories as:

$$\text{PET} = \min_{\substack{1 \leq k \leq N \\ 1 \leq \ell \leq M}} \{|t_\ell - t_k| : B_1(t_k) \cap B_2(t_\ell) \neq \emptyset\} \quad (1)$$

That is, PET is the smallest absolute time difference between any pair of timestamps at which the two objects' occupancy regions intersect. The PET can be extended to consider multiple conflict points between the same two trajectories. If $\text{PET} = 0$, the two vehicles occupied the same space at the same time and a direct conflict occurs. If $\text{PET} > 0$, the vehicles passed through the same region at different times. A small PET indicates a near-miss with elevated safety risk. If no intersection is found across any pair (t_k, t_ℓ) , the PET is undefined or may be considered infinite.

The challenge is to calculate the PET for all objects over long time horizons. PET calculations are increasingly provided in some 3rd party cloud service products; here we directly

compute them from the raw trajectories, considering the possible interactions of 10,000-20,000 objects per intersection per day over months of monitoring.

Figure 5 shows a heatmap of PET events at Location 2 on a single day (December 12, 2024). In the figure, High event severity is assigned if one of the road users is a Person (note: if both road users are a Person, then no PET event is generated).

B. Out of crosswalk trajectories

Through the included boundary definitions, the crosswalk and sidewalk zones are indicated across the corridor. The high-precision distance estimates from the lidar sensor permit analysis of persons walking outside of the crosswalk, independent of whether these represent PET violations. This degree of precision also enables fine-grained insights such as the degree to which individuals are cutting a corner or avoiding an obstacle.



Fig. 5. Post Encroachment Time (PET) for a representative day. The heatmap shows PETs for location 2 on December 12, 2024. “High” event severity is for events in which one of the trajectories involves a person. The size of the circle that indicates the event is nonlinearly inversely scaled by the time of the PET (e.g., a 0.2s PET is much larger than a 1.2s PET) to indicate the potential that it constitutes a near miss.

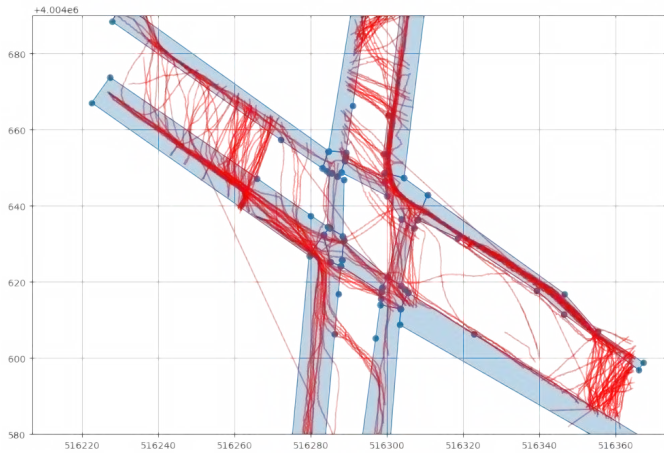


Fig. 6. Pedestrian trajectories that contain out-of-crosswalk segments greater than 10m in length, taken from 26th/Clarksville (location 1) on October 16, 2024.

Figure 6 shows a single day of out-of-crosswalk trajectories, defined as any trajectory containing a continuous segment of greater than 10 meters outside of a crosswalk or sidewalk zone. The figure displays the entirety of such trajectories, including the portions inside of zones. At this location, frequent crossings outside of the crosswalks display clustered behavior. Indeed, two convenience stores are located at the northeast corner of the intersection and generate frequent crossings of both roadways; a transit plaza is located at the southeast corner of the intersection and generates frequent crossings of Clarksville Pike on the eastern side of the intersection.

C. Turning movement examples

Standard traffic engineering turning movement counts can be computed by applying spatial analysis of trajectories intersecting defined roadway zones. Pairs of zones define a turning movement or a straight-through movement. Further specifying movement counts by hour of the day (e.g., as shown in Table II) or in 15-minute periods mirrors standard field data collection practices and can therefore be helpful



Fig. 7. Turning movement daily summary for Location 1 (26th/Clarksville) on October 16, 2024.

for planning or engineering purposes. A summary of a single day of movements at location 1 (26th/Clarksville) is shown in Figure 7.

Turning count movement data may indicate ‘lost’ vehicles, since vehicles that move into commercial parking lots and/or exit through a different path (or with less than 12m of trajectory distance) may not be counted as entering a different road segment. Likewise, this may mean that turning counts may sum to less than the total number of vehicle trajectories available on that day.

TABLE II
INTERSECTION MOVEMENT COUNTS AT 26TH/CLARKSVILLE BY HOUR OF DAY, SHOWN FOR MORNING PERIOD ON OCTOBER 16, 2024.

mvmt	hour	7	8	9	10	11	12	13
WB_th		187	221	249	297	360	392	376
WB_rt		9	13	15	24	20	19	16
WB_lt		2	7	5	12	13	9	9
EB_th		576	496	371	330	338	413	332
EB_rt		27	30	28	25	28	29	23
EB_lt		17	20	20	42	35	20	32
NB_th		2	3	4	3	6	5	3
NB_lt		27	17	32	36	30	40	18
NB_rt		14	14	7	19	22	21	19
SB_th		2	3	1	4	5	7	8
SB_lt		16	10	14	14	12	34	22
SB_rt		31	17	26	27	46	37	45

VII. DATA RELEASE

The data are available for download at [23]. A tutorial is provided from [21] that with step-by-step instructions to create the analysis environment in python, process trajectory files, and produce the plots in the paper. The data files provided in the public downloaded format are extracted from the production database, with filtering applied as described in Section V, and delivered through binary python object files.

Analysis performed in the paper is based on synchronous calls to the database, rather than using the downloaded files, for ease of cross-day analysis through advanced queries in the production of the summary data used to synthesize Figure 4. Thus to reproduce the summary data for this plot will require additional software to iterate over each of the files, if the publicly available data are used instead. The data used to produce the plots are included in [23].

VIII. FUTURE WORK

Additional analysis and discoveries by the community will be unlocked by this dataset. Correlation of pedestrian action near large vehicles (buses), and bicycles, can provide insights into multi-modal usage.

In Stage II of this project, 110 new intersections and mid-block areas will receive sensor installations, predominantly along the Nolensville Pike corridor in Nashville and major connector streets. A separate installation is planned for Broadway in downtown Nashville, with five sensors locations (1st Ave through 5th Ave) providing complete coverage of the roadway. This installation will support public safety by alerting traffic operations managers and law enforcement with pedestrian volume warrants a partial or full closure of the street. All future stages of the project will seek integration with signal controllers, which can enable new control applications such as pedestrians phase extension.

We expect that future data at the increased scale will be released in a similar manner as this dataset.

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